Moving beyond the average: environmental stochasticity as a driver of mammalian movement

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Abstract

The range size of an animal is often used of as a proxy for the animal's overall needs, inlcluding access to food, water, shelter, and mates. Range sizes also depend on several latent variables, such as vagility, competition, predation, foraging behaviour, patch depletion, and reproductive behaviour. Animals' range sizes and mass follow a well-documented power relationship with exponent near 3/4, which suggests that animals' range sizes scale with their average metabolic needs. While this simple power relationship can explain the great majority of the deviance in range size, most of the remaining deviance can be explained by accouning for resource abundance and stochasticity (unpredictability). Using an intercontnental dataset of 6.3 million locations of 1054 terrestrial mammals across 34 species, we present empirical evidence that, on average, mammals reduce their range size with increasing resource abundance and increase their range with increasing resource stochasticity. Additionally, the two variables interact, such that higlhy predictable or stocastic environments result in stronger range contractions as resources increase. Conversely, environments with very scarce or abudant resources cause mammals to respond to resource stochasticity in a concave response. When resources were scarce, ranges were smallest at intermediate stochasticity and largest when in higly predictable or stochastic environments. When resources were abundant, the relationship flipped: ranges were smallest at low and high stochasticity, and largest at intermediate stochasticity. Thus, environmental stochasticity shapes how mammals respond to resource availability, and accounting for environmental stochasticity is essential for understanding how mammals respond to dynamic environments. As environmental stochasticity increases due to climate change and anthropogenic landscape transformation (e.g., tree harvesting, mining, farming, urban expansion), moving beyond average conditions and accounting for resource stochasticity will provide a more complete picture of the drivers of mammalian movement. NEED A CLOSING SENTENCE OR TWO.

1 Introduction

The drivers of animal movement are many and complex, including searching for resources, avoiding unwanted competition, and escaping danger[1]. However, this complexity implies that the range size of an animal can used as a proxy for the animal's needs[2,3]. This hypothesis is supported by the fact that range size and body size scale with an exponent near $\frac{3}{4}$ across several animal classes[4], which matches the scaling of metabolic needs with body size[5]. Thus, body size alone can explain most of the variation in range size across species[4], but other drivers are also important. Range size often decreases with resource abundance, since greater average resource abundance allows animals to remain within smaller areas[6]. In addition, the variation around the average, i.e., resource stochasticity, also impacts how and where animals move[7].

To understand how resource abundance and stochasticity interact, it helps to think of resources as a random variable[8]. Each patch in a habitat is then a realization of the random variable, with the typical value of a patch is given by its resource abundance, while the patchiness of the environment and the stability of individual patches are determined by resource stochasticity. Importantly, the two parameters interact: at low resource abundance, increasing resource stochasticity transforms a homogeneous and predictable habitat (e.g., a steppe with some greener areas) into a heterogeneous and unpredictable habitat that has very rare but lush patches (e.g., a desert with scattered oases). Increasing environmental stochasticity at high resource abundance results in a similar but opposite habitat, since resources increase across the whole habitat with the exception of rare 'dead' patches.

Since each of the four habitat types will have different carrying capacities and selective pressures, species will develop different strategies and competitive behaviours. Resource-poor habitats may promote territoriality and caching[**wolverine_ref_andrea?**] or nomadism[**mongolian_gazelle?**], while resource-rich habitats may have low rates of competition[**ref?**]. Inversely, predictable habitats may favor species coexistence[**ref?**] and

diversity[9], while stochastic habitats may promote despotic competition that forces low-rank individuals towards low-quality patches[10]. Understanding the combined effects of resource abundance and stochasticity thus requires accounting for both variables concurrently[8], along with any other confounding variables, such as competitive pressure and resource depletion[11]. Thus, telemetry data of wild animals is essential for including such confounds and sources of variance in the estimates.

Using a large intercontinental telemetry dataset with 6.3 million locations of 1054 animals from 34 distinct species (Fig. 1), we present provide estimates of the effects of resource abundance and stochasticity on the range sizes of terrestrial mammals'. We use long-term estimates of spatiotemporal mean NDVI and variance in NDVI as proxies for resource abundance and stochasticity, respectively. To account for the nonlinear effects of the two variables, including nonlinear interaction effects, we model the relationships using a Generalized Additive Model[12] (GAM). Specifically, we use a hierarchical GAM[13] for location and scale[12,14,15] to account for similarities across conspecifics and estimate common trends across species. The model accounts for: the partial effects of mean and variance in NDVI for each species, partial effects of the common partial effects across species, and a tensor product interaction term to account for the nonlinear interaction …

https://pubmed.ncbi.nlm.nih.gov/23020517/

- Migrating animals will have larger range sizes and smaller EDF values, and they will also be more susceptible to interannual variation. These two points work well together because they imply we should estimate changes in the mean as smooth changes that occur across years rather than among years, which means i should keep the k for years relatively low (e.g., (2025 - 1981) / 4 = 11 10)

human activity is increasing stochasticity and conseuqnetly HR size, but it's also constraining movement capacities (Tucker paper) and attracting to urban traps

follow up after main figure: - changes in variance over time: spatial needs will increase in the future, exacerbating issues - increases in variances => increases in movement rates

=> increases in energy expenditure, which increases HR size further? what if animals can't achieve an energetic balance? - increases in variances => increases in movement rates => increases in encounter rates => how will this affect trophic structures and predation rates? what if prey are forced to alter the predator/prey dynamics to try to survive? - increases in env var can have strong cascading effects that then increase the var of other processes and lead to a series of strong cascades and maybe collapse

difficulties with estimating effect of variance:

- mean and variance are correlated
- variance effect seems to be weaker than effect of mean
- importance of scale (rasters are at ~5 km res)
- temporal vs spatial variance
- variance in NDVI may not be representative of environmental variance: a light in a boreal forest snow can change NDVI substantially without impacting nutrient availability

papers to use for structure: - jetz scaling of animal space use (https://pubmed.ncbi.nlm.nih.gov/15472074/) - tucker moving in the anthropocene (https://www.science.org/doi/10.1126/science.aam9712) - the search behavior of terrestrial mammals (https://www.biorxiv.org/content/10.1101/2022.12.31.521874v1)

artificial feeding can alter community structures in ways that are not reflected by NDVI: https://doi.org/10.1002/ecs2.70368

variance has u-shaped effect at low means: some variance at low means is beneficial because it allows to move across patches as they deplete, if animal can find good patches.

useful refs: - uses ndvi variance: https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348 -

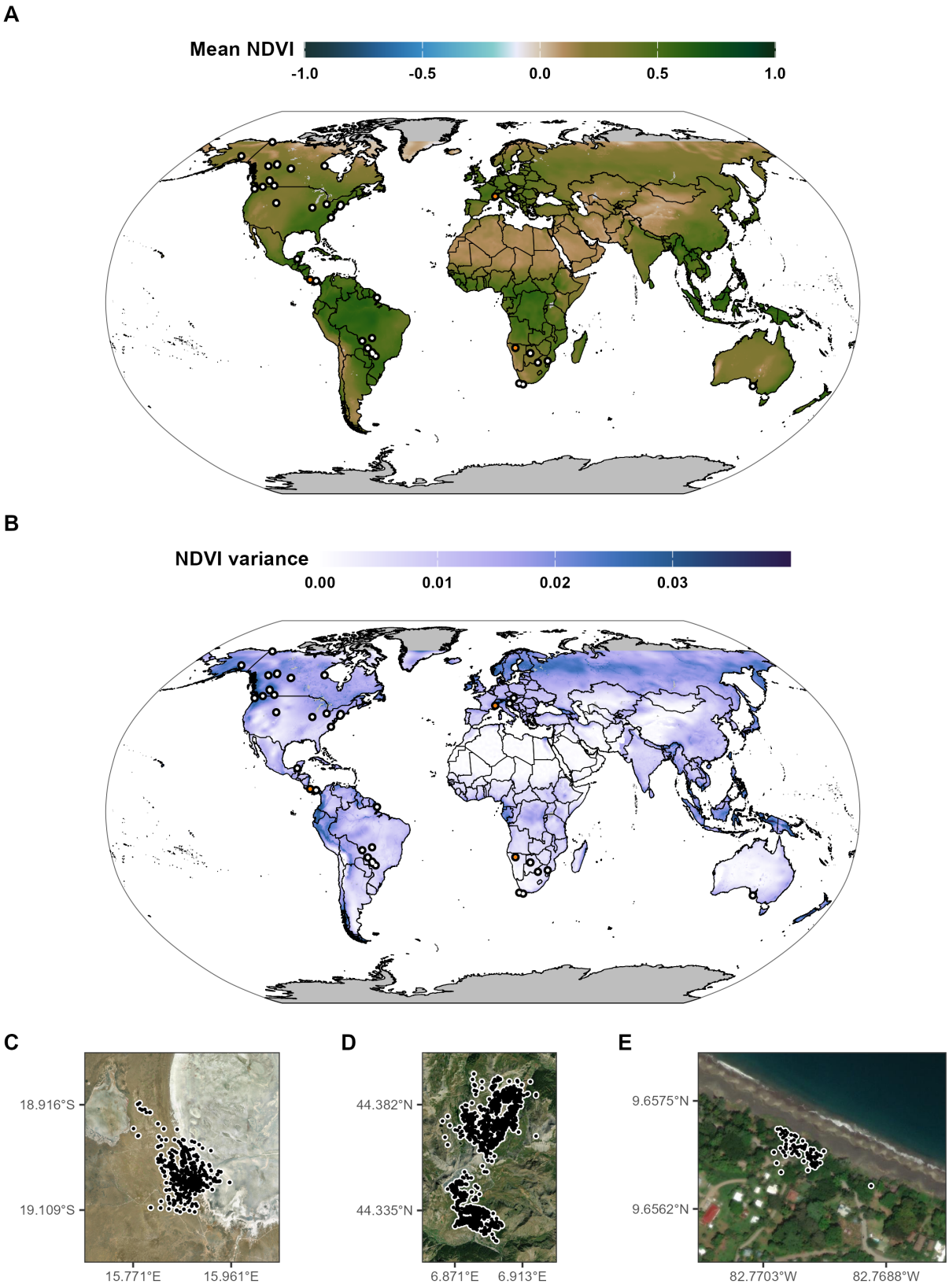


Figure 1: Maps of the rasters of mean NDVI (A) and variance in NDVI (B) used as proxies for mean and variance in resource abundance. The bottom three panels show examples of environments with: (C) low mean NDVI and variance (*Canis mesomelas* near the Etosha Pan in Namibia), (D) intermediate mean NDVI and variance (*Capra ibex* in the Alps), and (E) high mean NDVI and variance (*Choloepus hoffmanni* in Costa Rica).

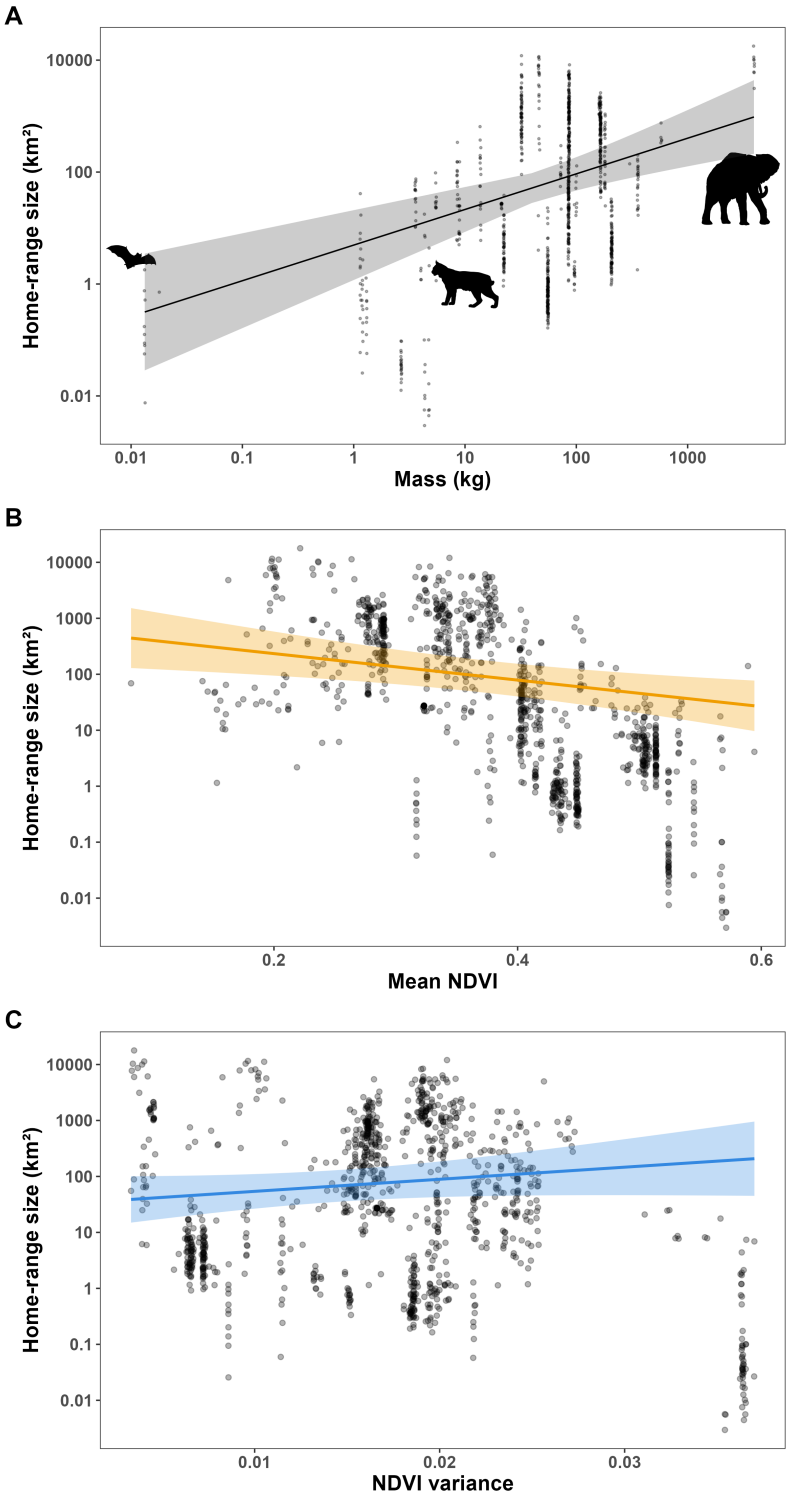


Figure 2: Partial effects of body size (A), mean NDVI (B), and NDVI variance (C) on home range size. Y axes are on the log scale. Masses in panel A are jittered horizontally to aid in visualization, but each species has a single value for mass. The animals in the bottom right corner of panel C have masses in the bottom 8th percentile of the dataset (species: *Bradypus variegatus*, *Carollia castanea*, *Carollia perspicillata*, *Choloepus hoffmanni*, *Dasyprocta punctata*, *Nasua narica*).

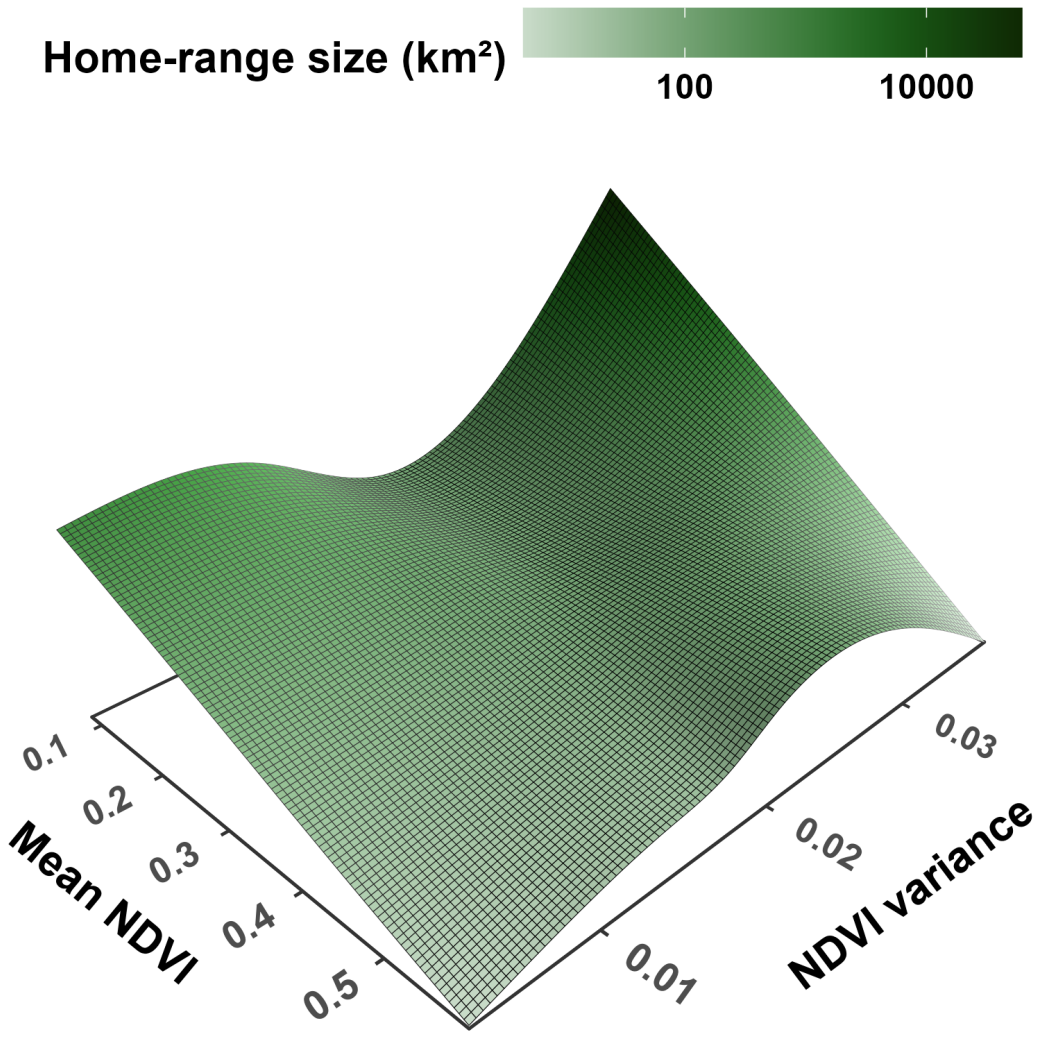


Figure 3: Interaction effects of mean NDVI and variance in NDVI on the home range size of a 85-kg mammal. Home range size is on the log scale in both the z axis and the fill scale.

2 Materials and Methods

for buffalo data, cite:

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